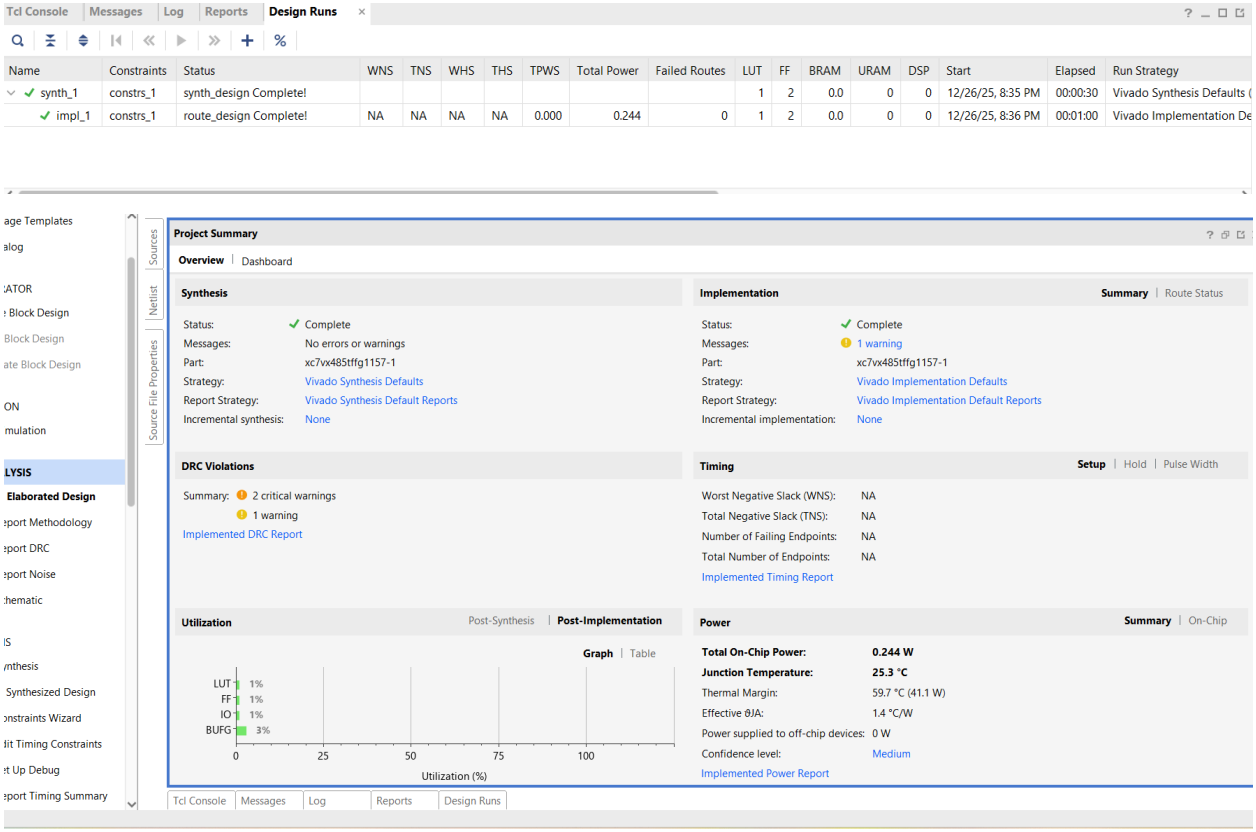
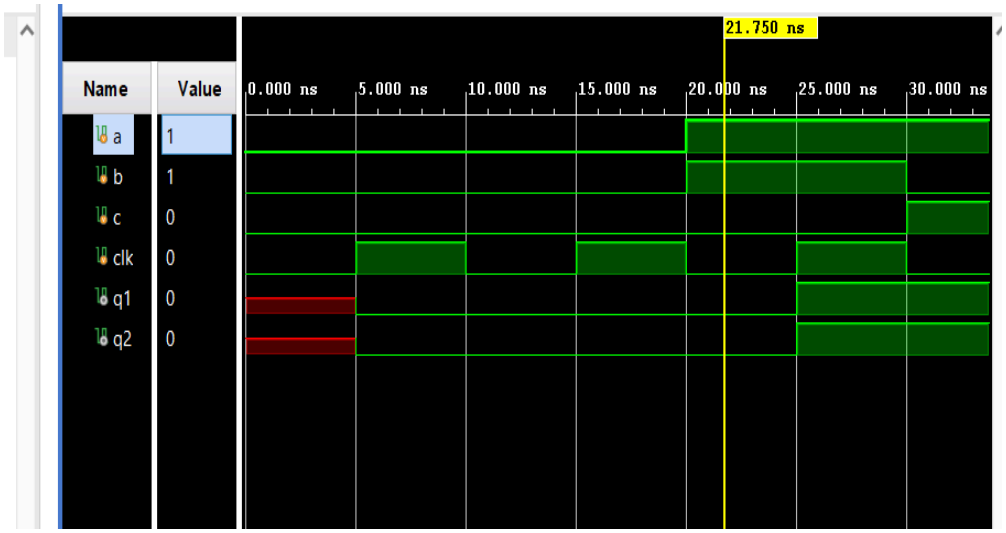
Report
Setayesh saeidi - 40131851

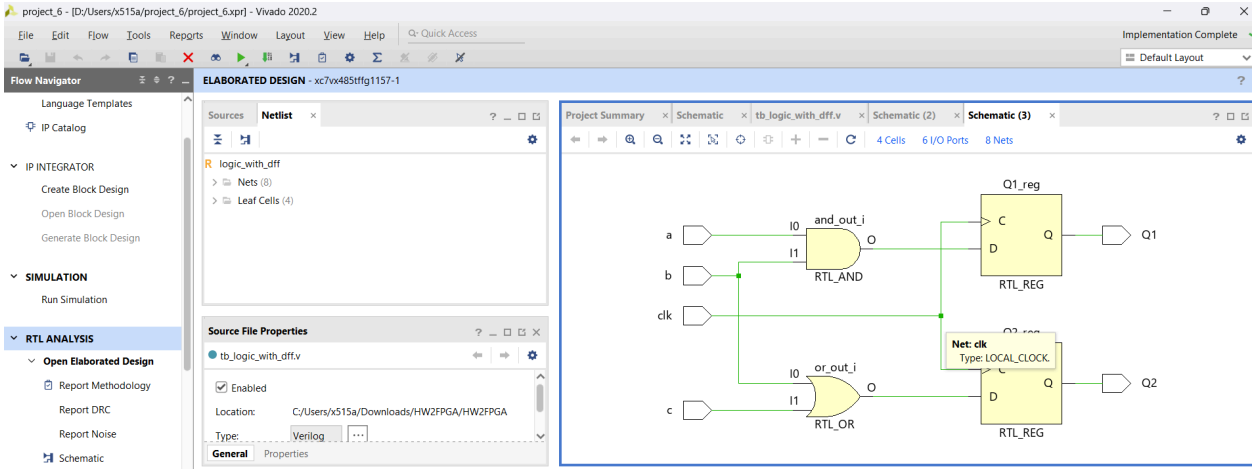
Q1.
Synthesis :



Test bench res :



Circuit schematic:



Q2.

Circuit Description:

In a Ripple Carry Adder, the carry output (C_{out}) of the Least Significant Bit (LSB) stage is connected directly to the carry input (C_{in}) of the next significant bit. This means the 4th adder cannot finish its calculation until the 3rd adder is done, which cannot finish until the 2nd is done, and so on. The carry signal "ripples" through the chain.

Critical Path Calculation:

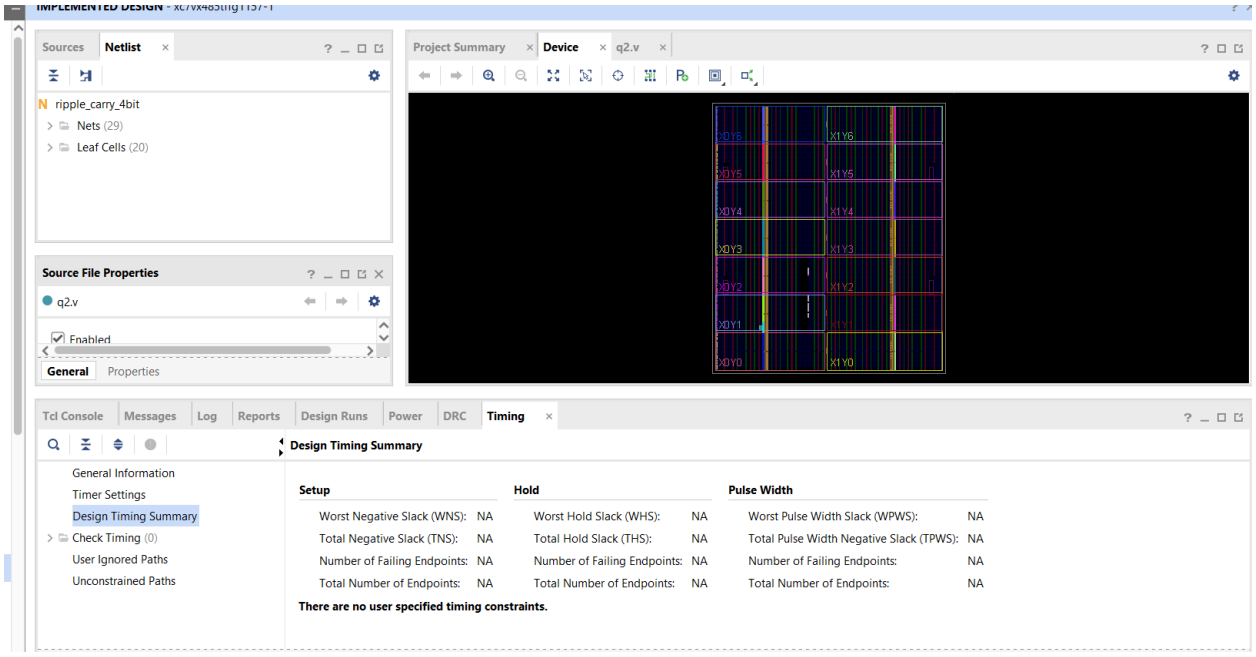
- **Let T_{carry} be the delay for a Full Adder to generate a Carry Output.**
- **Let T_{sum} be the delay for a Full Adder to generate a Sum Output.**
- **The Critical Path is the path that the carry signal takes from the LSB input (A_0, B_0) all the way to the MSB sum (S_3) or final carry (C_{out}).**

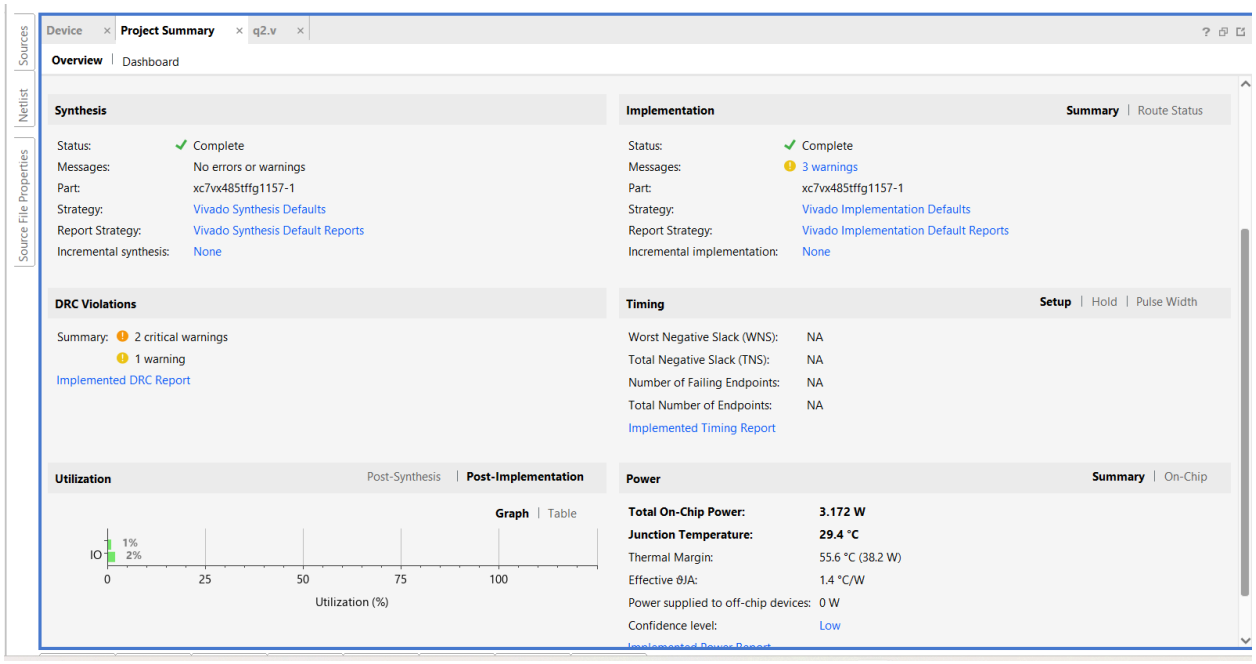
For this 4-bit adder:
The critical path goes through:

1. **FA0 (Generate Carry)**
2. **FA1 (Propagate Carry)**
3. **FA2 (Propagate Carry)**
4. **FA3 (Generate Final Carry or Sum)**

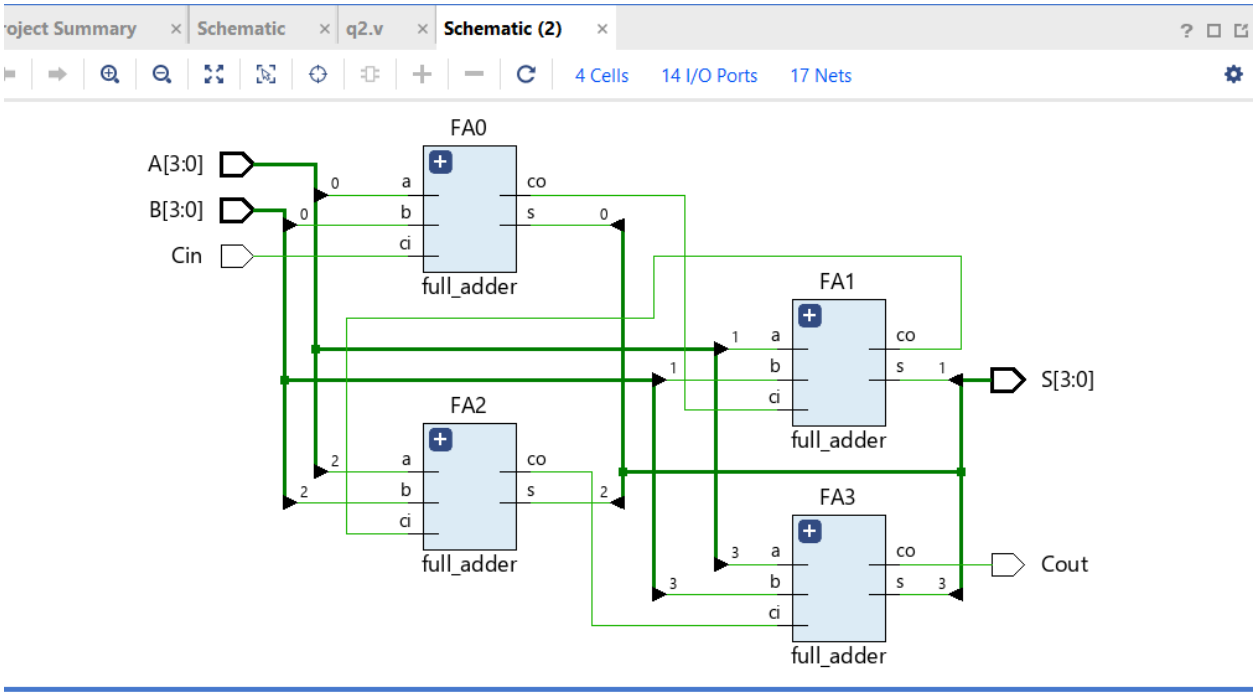
This linear delay ($O(N)$) makes the Ripple Carry Adder the slowest type of adder compared to parallel prefix adders (like Carry Lookahead), but it has the lowest area (smallest size).

Success synthesis:

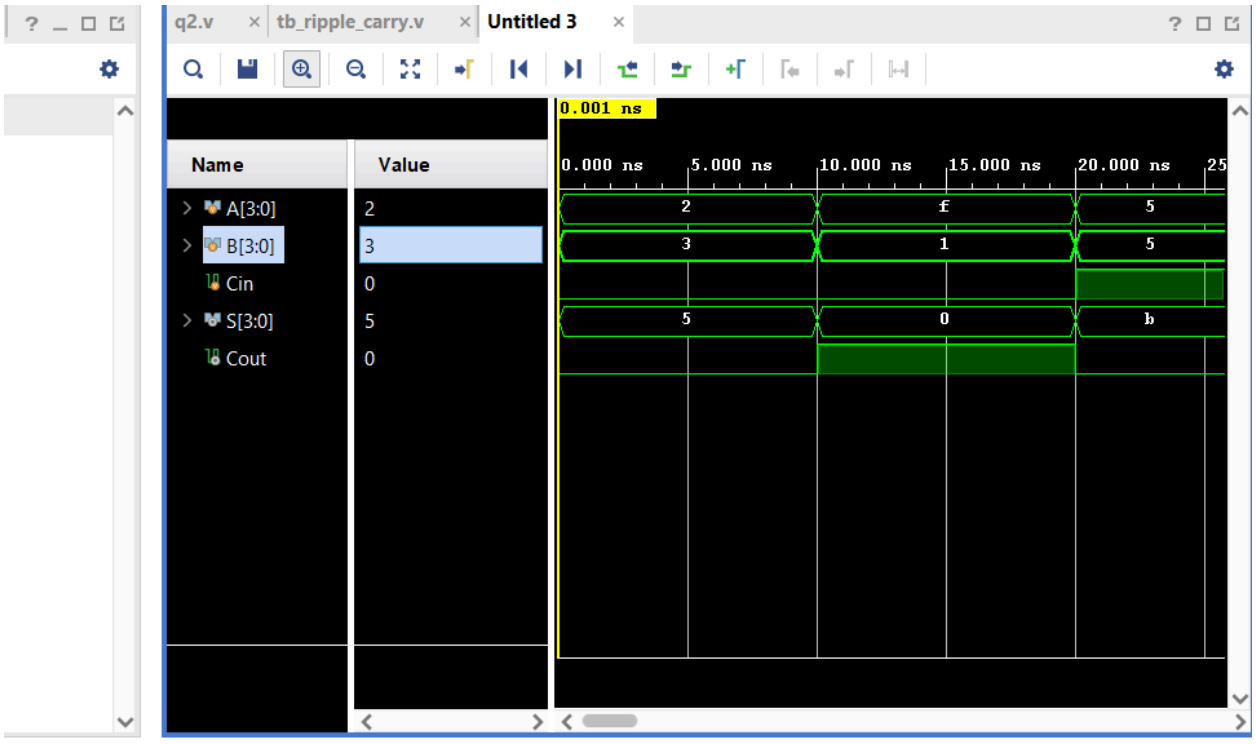




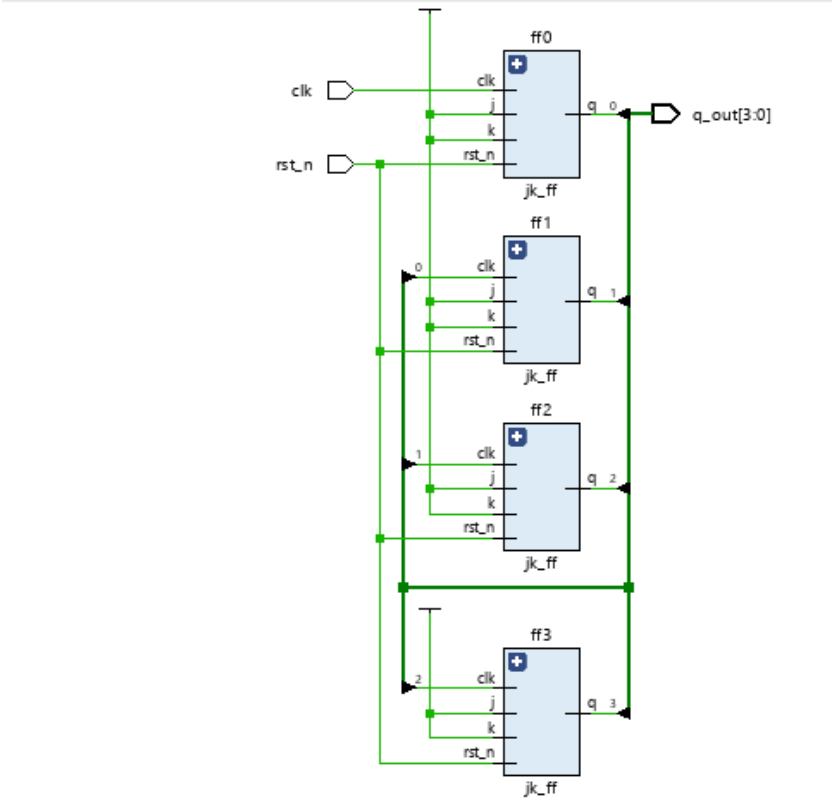
schematic:

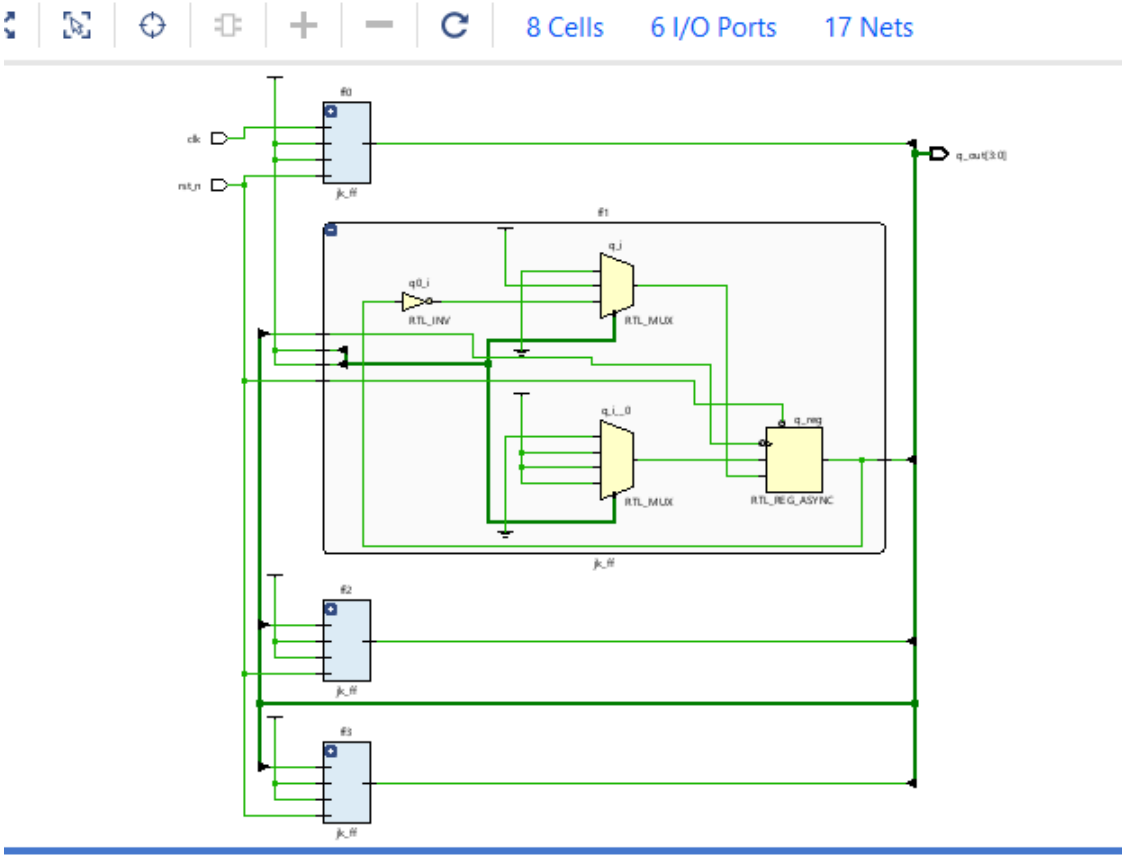

oject Summary
Schematic
q2.v
Schematic (2)
4 Cells
14 I/O Ports
17 Nets
FA0
A[3:0]
B[3:0]
Cin
a
b
ci
co
s
full_adder
FA1
FA2
FA3
S[3:0]
Cout

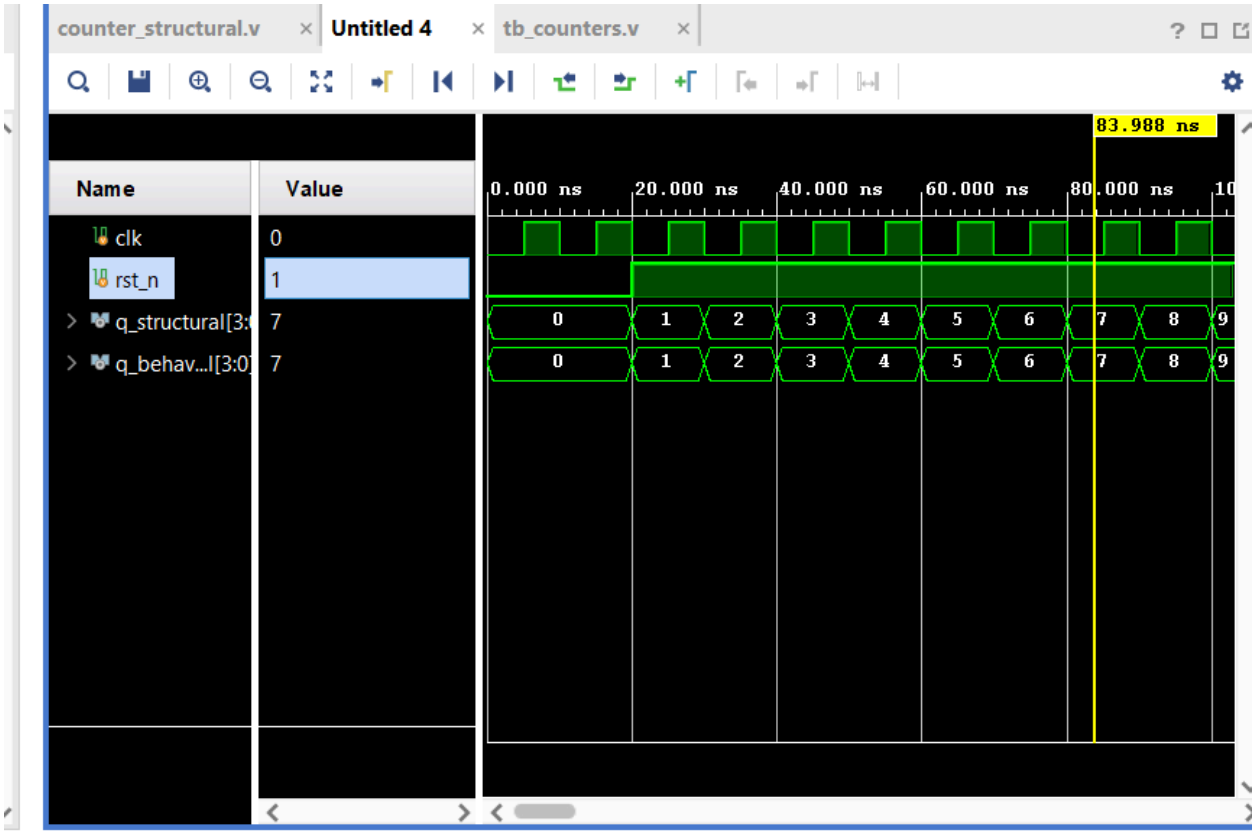
Test bench :


q2.v
tb_ripple_carry.v
Untitled 3
Name
Value
A[3:0]
B[3:0]
Cin
S[3:0]
Cout
2
3
0
5
0
0.001 ns
0.000 ns
5.000 ns
10.000 ns
15.000 ns
20.000 ns

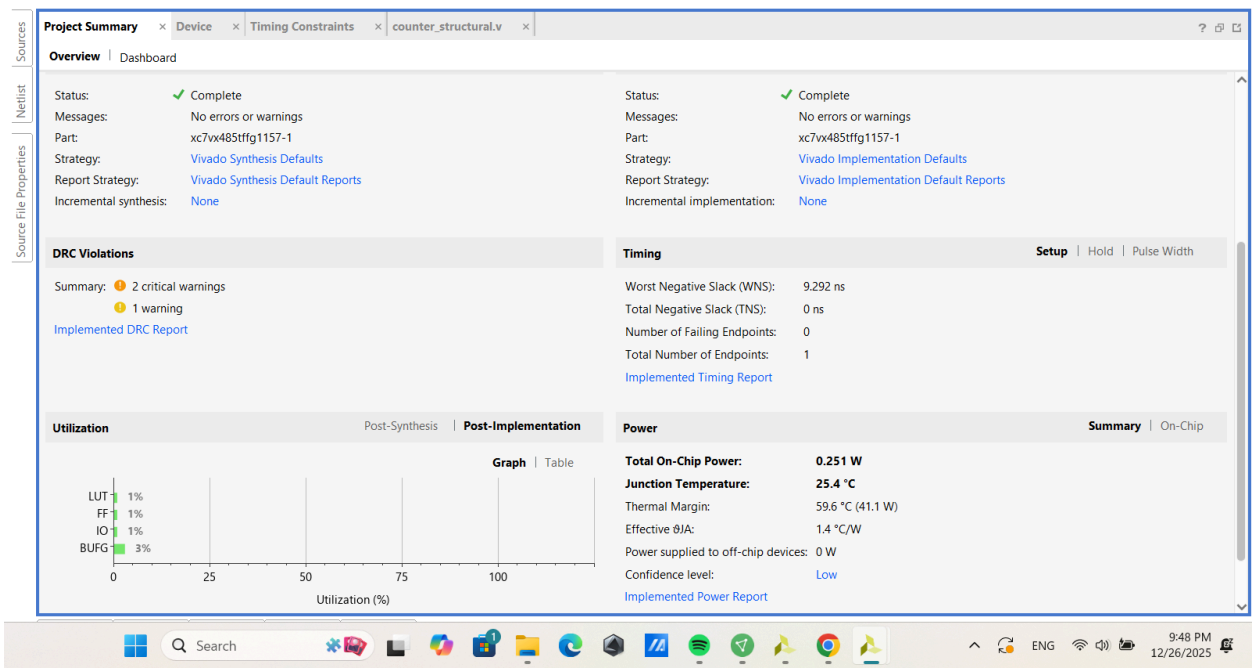
q3:structual

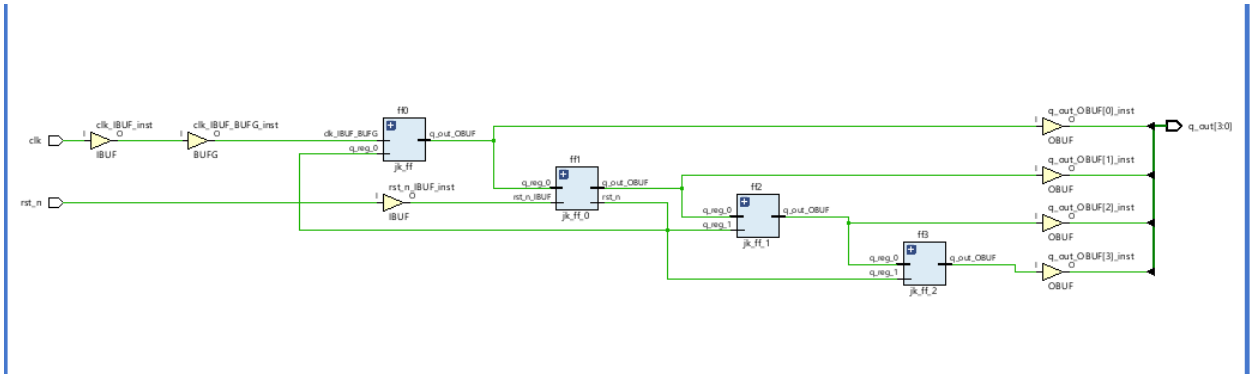


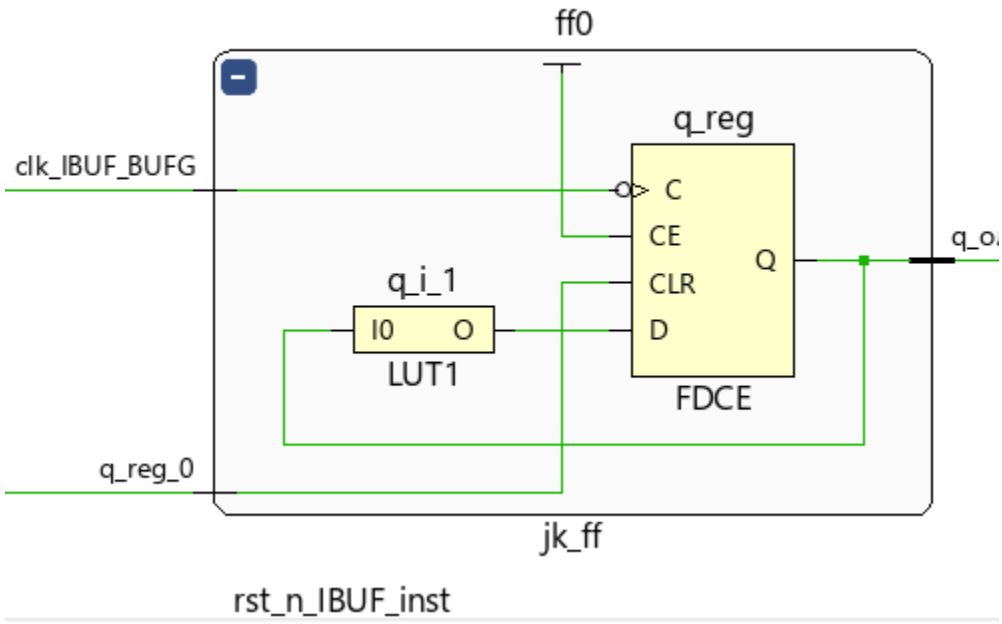


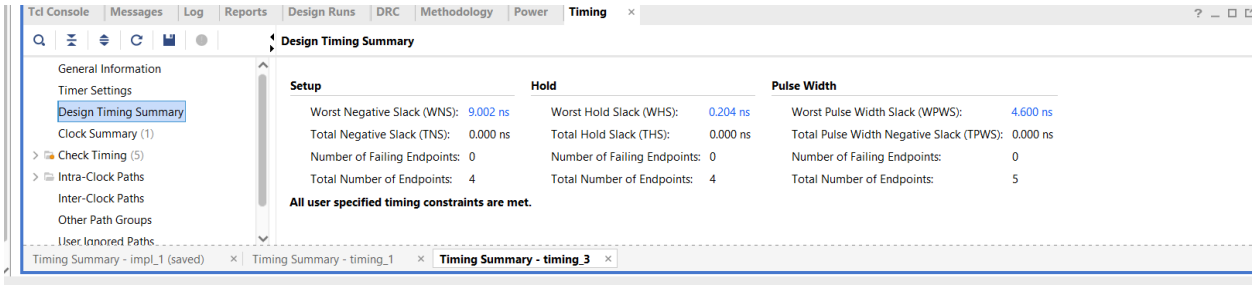

counter_structural.v
Untitled 4
tb_counters.v
Name
Value
clk
0
rst_n
1
q_structural[3:
7
q_behav...l[3:0]
7
83.988 ns
0.000 ns
20.000 ns
40.000 ns
60.000 ns
80.000 ns
0 1 2 3 4 5 6 7 8 9
0 1 2 3 4 5 6 7 8 9

success:


Project Summary
Device
Timing Constraints
counter_structural.v
Overview
Dashboard
Status: Complete
Messages: No errors or warnings
Part: xc7vx485tffg1157-1
Strategy: Vivado Synthesis Defaults
Report Strategy: Vivado Synthesis Default Reports
Incremental synthesis: None
Status: Complete
Messages: No errors or warnings
Part: xc7vx485tffg1157-1
Strategy: Vivado Implementation Defaults
Report Strategy: Vivado Implementation Default Reports
Incremental implementation: None
DRC Violations
Summary: 2 critical warnings
1 warning
Implemented DRC Report
Timing
Setup | Hold | Pulse Width
Worst Negative Slack (WNS): 9.292 ns
Total Negative Slack (TNS): 0 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 1
Implemented Timing Report
Utilization
Post-Synthesis | Post-Implementation
Graph | Table
LUT 1%
FF 1%
IO 1%
BUFG 3%
0 25 50 75 100
Utilization (%)
Power
Summary | On-Chip
Total On-Chip Power: 0.251 W
Junction Temperature: 25.4 °C
Thermal Margin: 59.6 °C (41.1 W)
Effective θJA: 1.4 °C/W
Power supplied to off-chip devices: 0 W
Confidence level: Low
Implemented Power Report
Search
ENG
9:48 PM
12/26/2025







Behavioral 👆

1. RTL Schematic Analysis

- **Structural Design (b):**
  - **Appearance: The schematic displays four distinct Flip-Flop blocks. Crucially, you will see the output wire**

of one block connected directly to the clock pin of the next.

- **Complexity: It looks like a "chain". It visually represents the dependency of bit N on bit $N-1$.**

- **Behavioral Design (c):**
  - **Appearance: The schematic displays a hardware Adder (or incrementer) block feeding back into a single block of 4 Registers (FDCE).**
  - **Connectivity: All 4 registers are connected to the same global `clk` line. This is a "Synchronous" design.**

2. Maximum Clock Speed (Frequency)

- **Structural (Ripple) - SLOW:**
  - **In design (b), the clock signal must propagate through each flip-flop.**
  - **Delay Formula: $T_{total} = 4 \times T_{clk-to-q}$.**
  - **The MSB (Bit 3) does not settle until the signal has rippled through Bits 0, 1, and 2. This drastically reduces the maximum operating frequency.**
  - ***Glitches:* Intermediate states (like $0111 \to 1000$) cause temporary false outputs during the ripple transition.**
- **Behavioral (Synchronous) - FAST:**
  - **In design (c), all flip-flops trigger simultaneously on the same system clock edge.**
  - **Delay Formula: $T_{total} = T_{logic} + T_{setup}$.**
  - **Modern FPGAs (like Xilinx Artix-7) use dedicated Carry Logic (CARRY4) for the addition. This allows the**

counter to run at very high speeds (typically > 400 MHz).

3. Resource Utilization (Xilinx Chip)

- **Structural (b):**
  - **Resources: Uses 4 Flip-Flops.**
  - **Clock Routing: This is the critical flaw. FPGAs have dedicated "Global Clock Trees" (BUFG) designed for low skew. Design (b) uses the *output* of a flip-flop as a clock. This forces the tool to use general logic routing for clocks, which is slow and introduces "Clock Skew". Vivado will issue severe warnings about "Generated Clocks".**
- **Behavioral (c):**
  - **Resources: Uses 4 Flip-Flops + 4 LUTs (Look-Up Tables) for the adder logic.**
  - **Routing: It uses the dedicated Global Clock Tree.**
  - **Conclusion: While it uses slightly more logic (LUTs) for the adder, it is the only professional way to design a counter on an FPGA.**

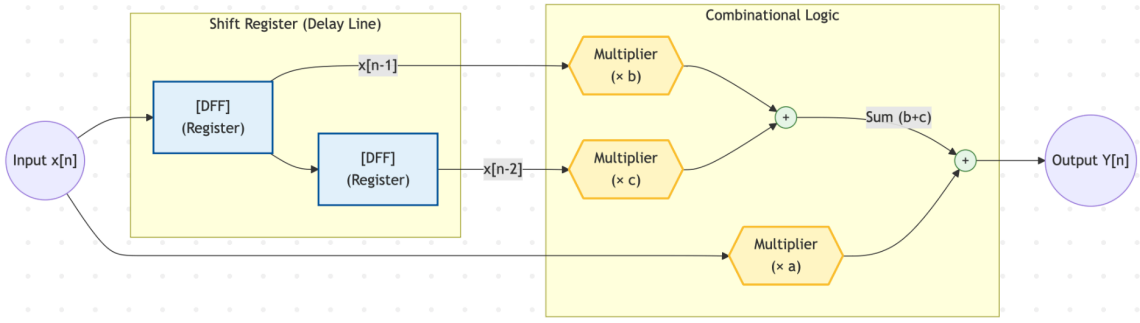
Feature	(b) Structural / Ripple	(c) Behavioral / Synchronous
Clocking	Asynchronous (Chained)	Synchronous (Shared)

Max Frequency	Low (Limited by chain delay)	High (Limited by 1 adder)
Clock Skew	High (Dangerous)	Zero (Ideal)
Glitches	Yes (Output is invalid during transition)	No (Clean transitions)
FPGA Preference	Poor (Avoid in FPGAs)	Excellent (Recommended)

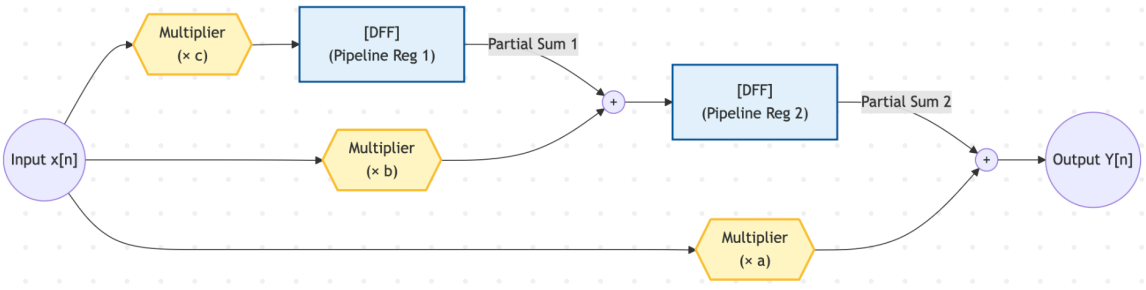
q4:

1.proposed circuit:

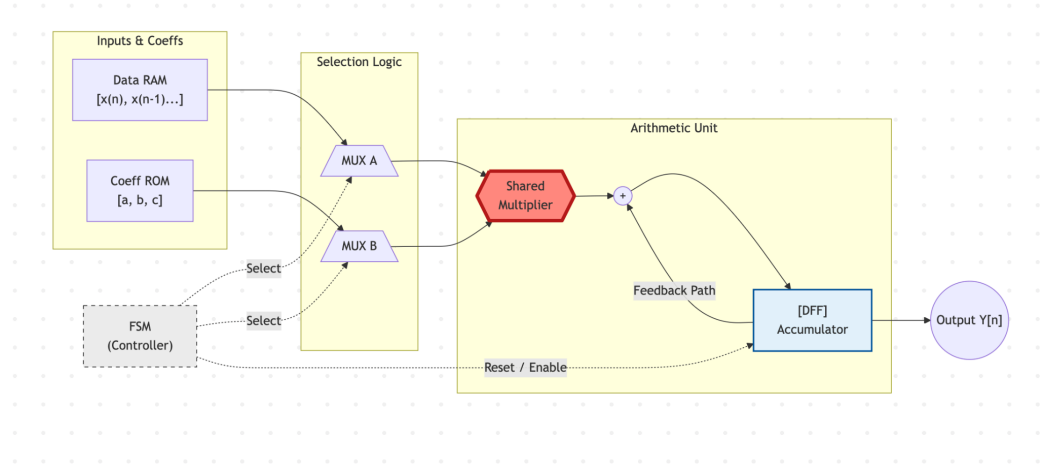
(a):Low Latency



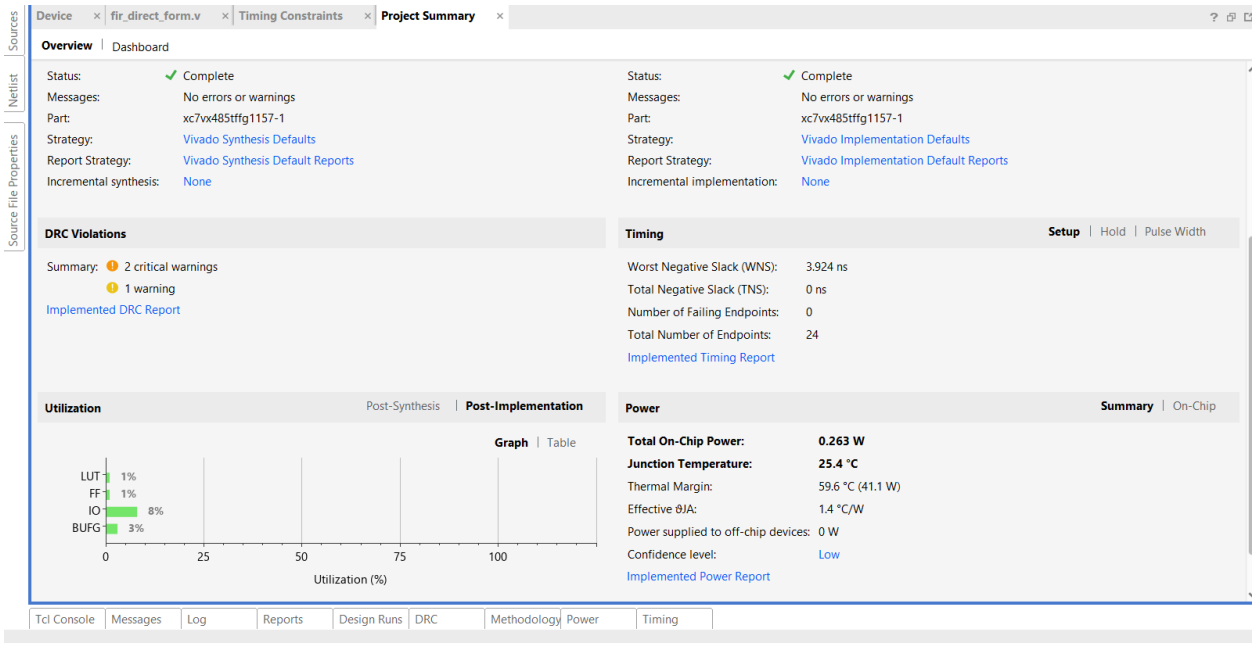
(b):High Throughput



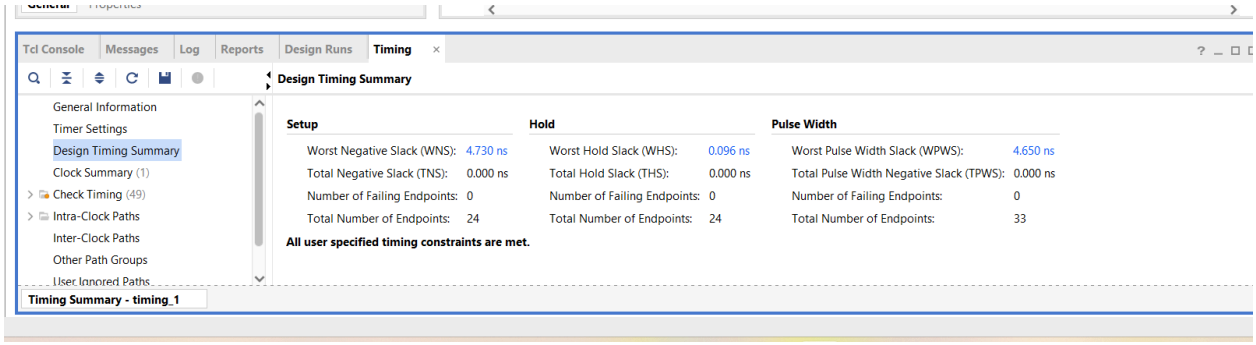
(c):Low Area



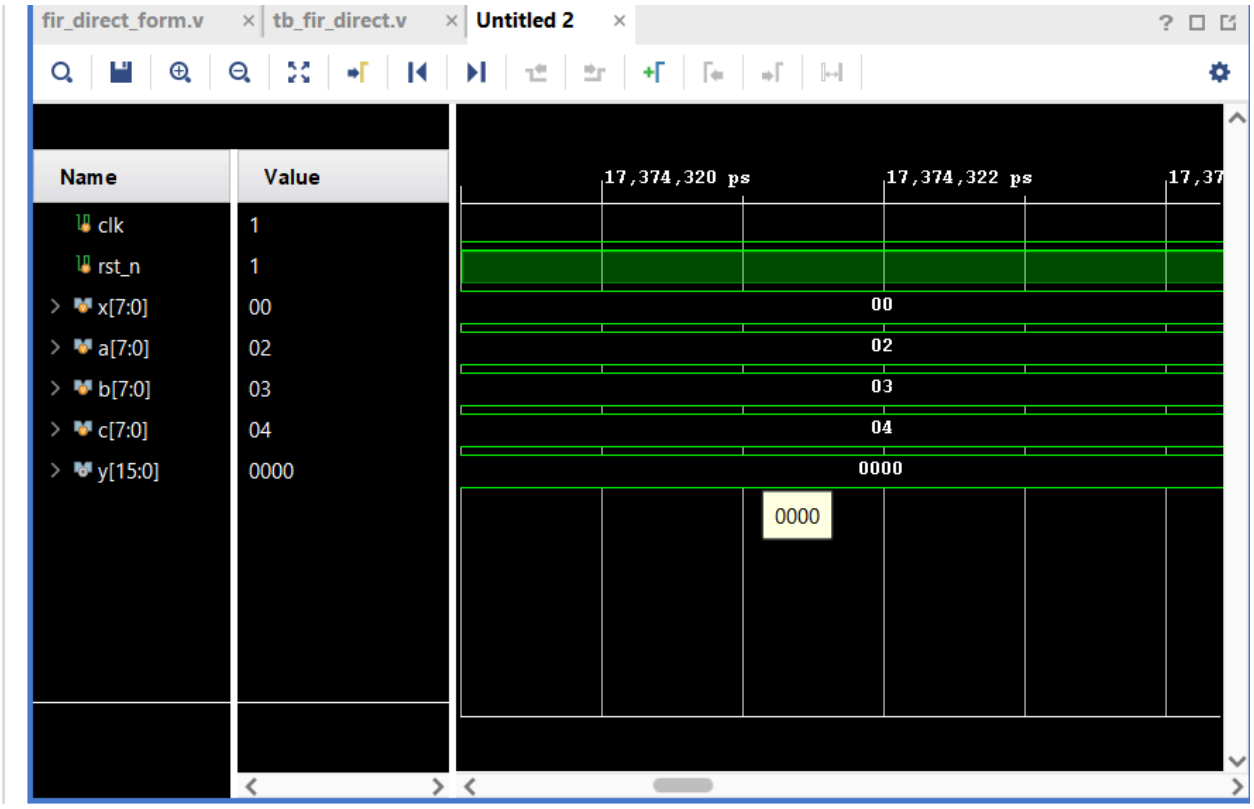
Low latency:
success



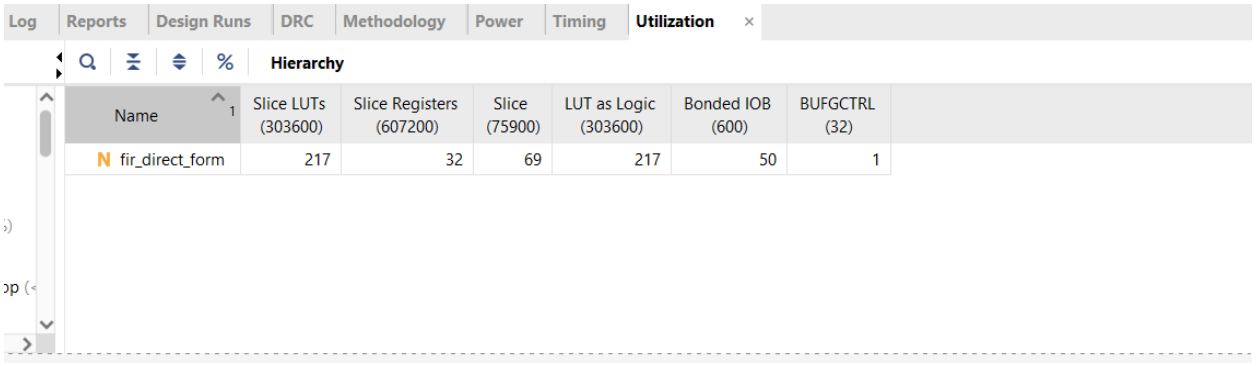
Timing report :



test bench :



Utilazation report :

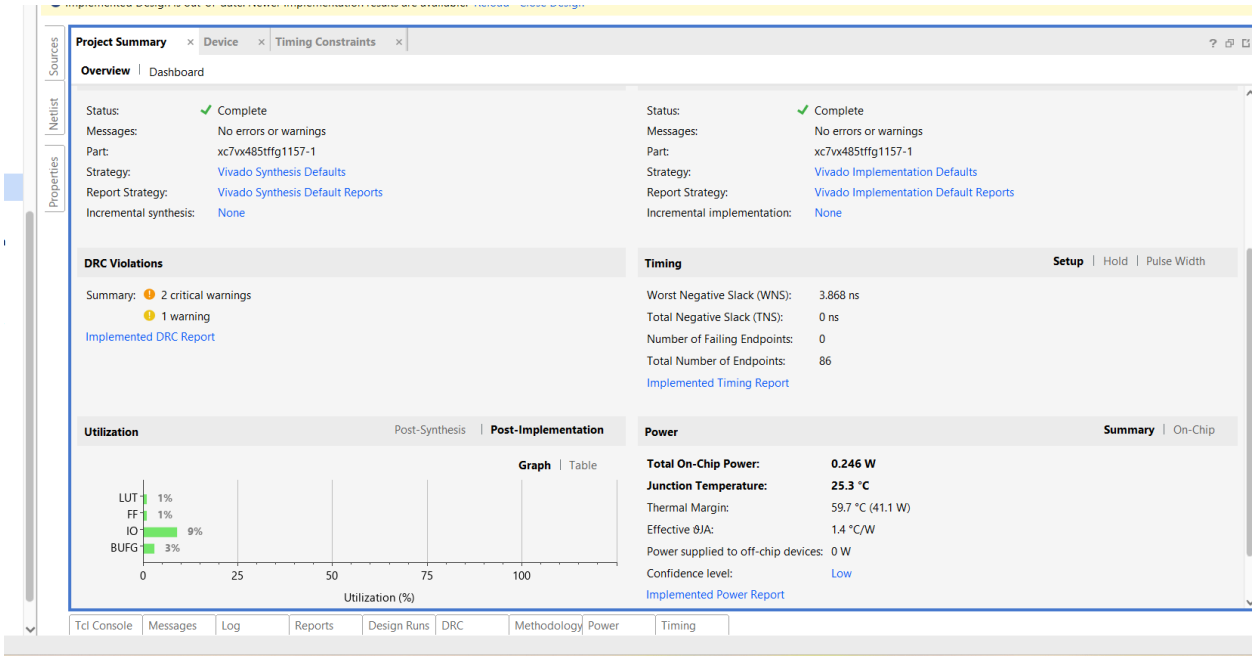


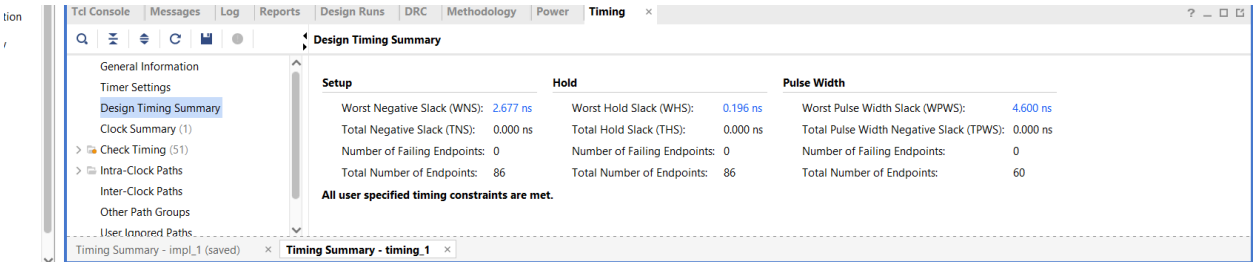
Utilization Post-Synthesis | **Post-Implementation**

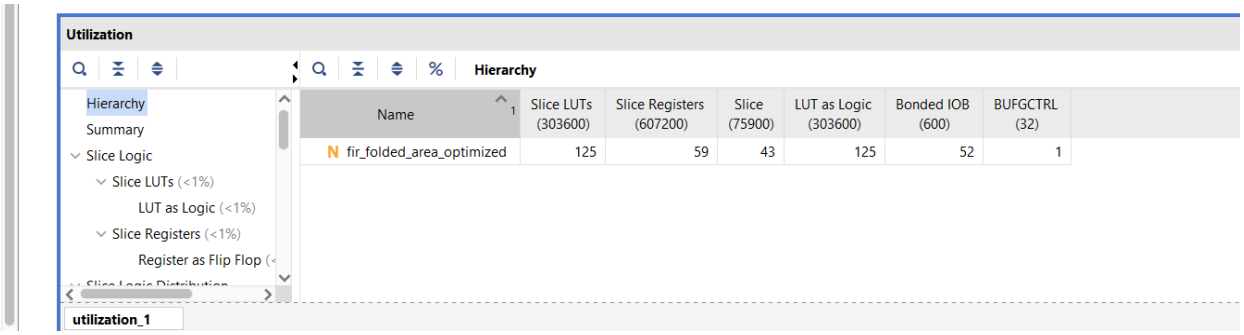
Graph | **Table**

Resource	Utilization	Available	Utilization %
LUT	217	303600	0.07
FF	32	607200	0.01
IO	50	600	8.33
BUFG	1	32	3.13

Folded ：





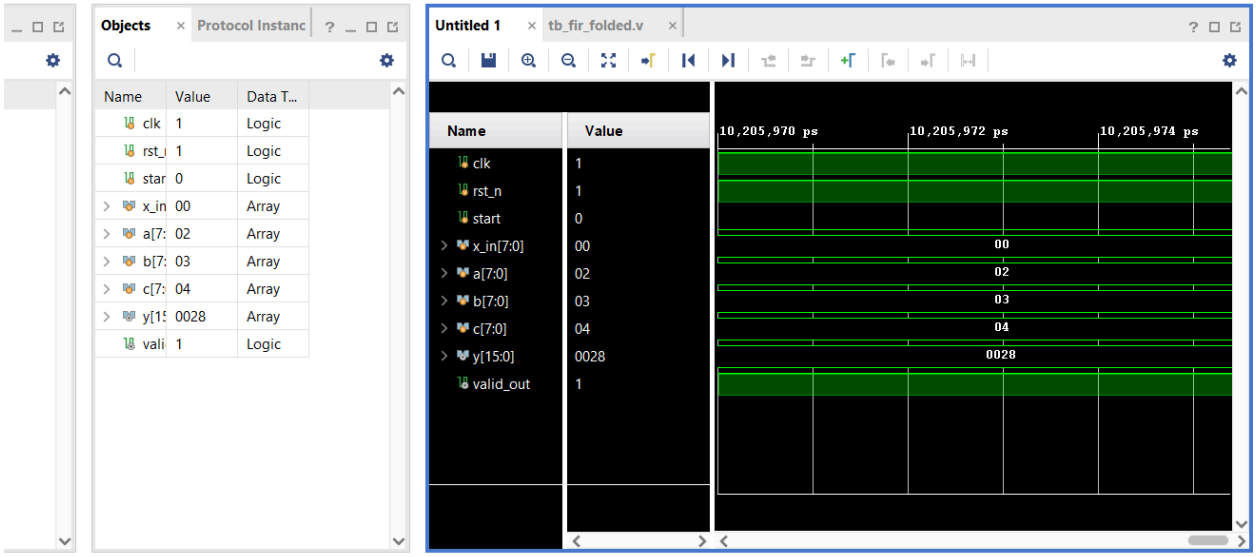


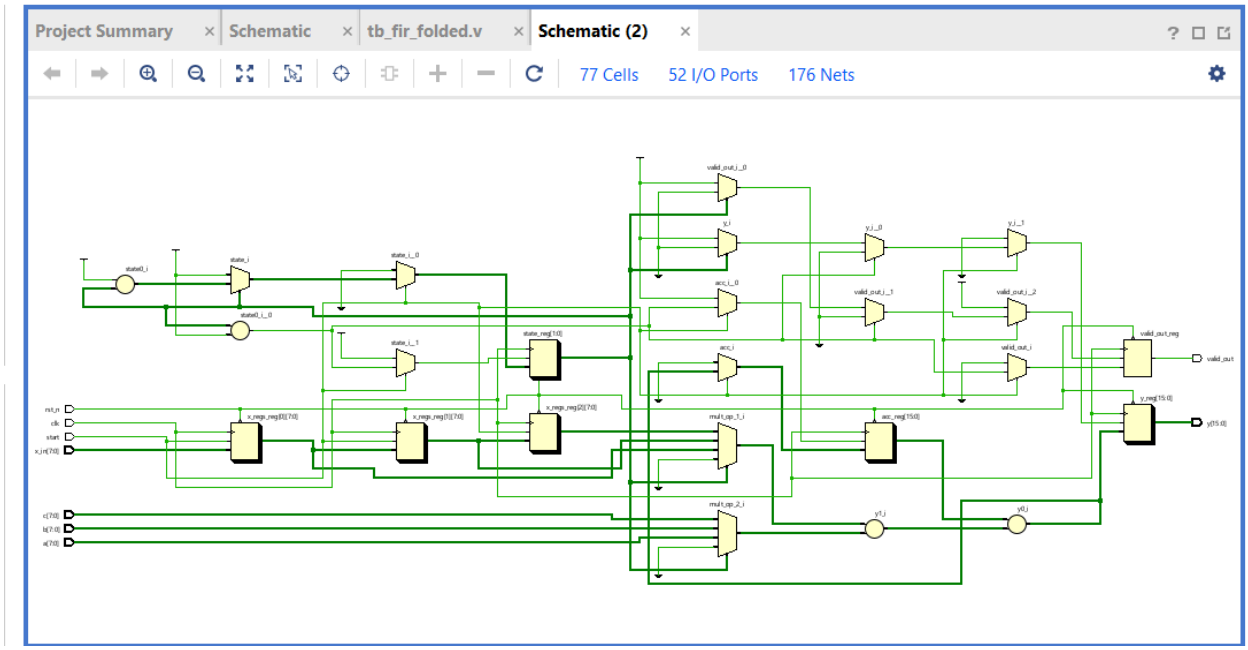
Utilization Post-Synthesis | **Post-Implementation**

Graph | **Table**

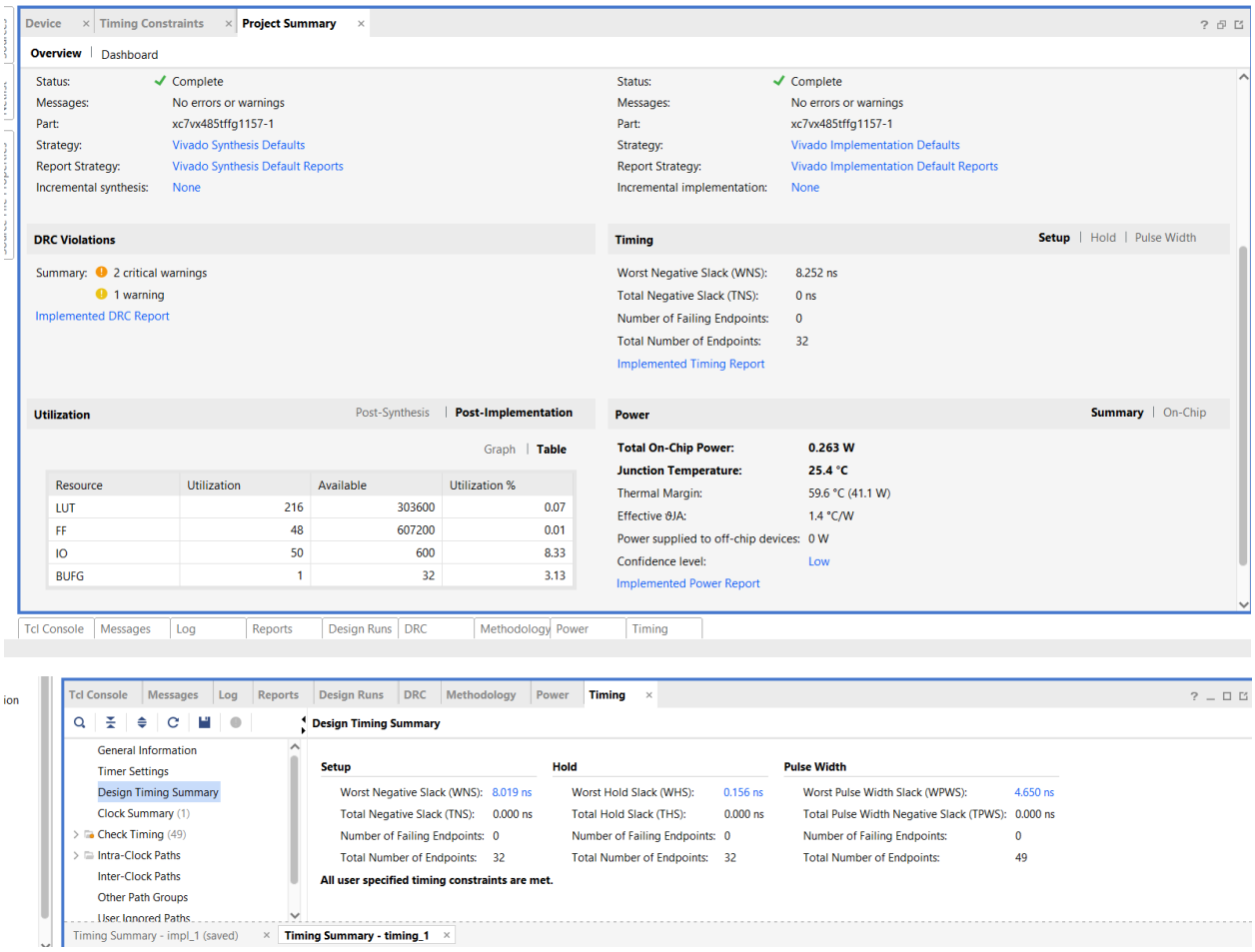
Resource	Utilization	Available	Utilization %
LUT	124	303600	0.04
FF	59	607200	0.01
IO	52	600	8.67
BUFG	1	32	3.13

Tcl Console | Messages | Log | Reports | Design Runs | DRC | Methodology | Pow

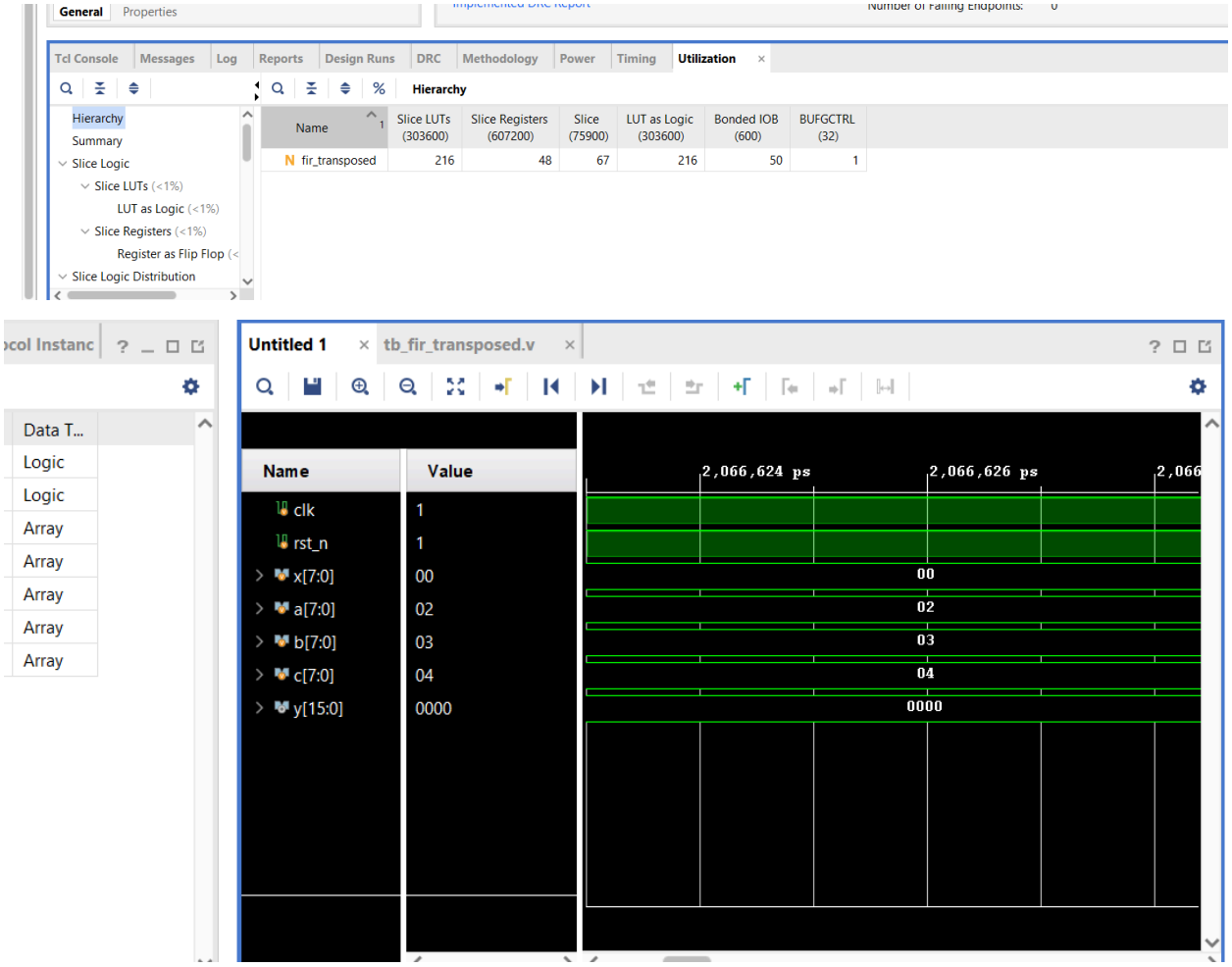

Objects
Protocol Instanc
Name
Value
Data T...
clk 1 Logic
rst_ 1 Logic
star 0 Logic
x_in 00 Array
a[7: 02 Array
b[7: 03 Array
c[7: 04 Array
y[15 0028 Array
vali 1 Logic
Untitled 1
tb_fir_folded.v
Name
Value
clk 1
rst_n 1
start 0
x_in[7:0] 00
a[7:0] 02
b[7:0] 03
c[7:0] 04
y[15:0] 0028
valid_out 1
10,205,970 ps
10,205,972 ps
10,205,974 ps
00
02
03
04
0028


Project Summary
Schematic
tb_fir_folded.v
Schematic (2)
77 Cells
52 I/O Ports
176 Nets

Transposed :


Device
Timing Constraints
Project Summary
Overview
Dashboard
Status: Complete
Messages: No errors or warnings
Part: xc7vx485tffg1157-1
Strategy: Vivado Synthesis Defaults
Report Strategy: Vivado Synthesis Default Reports
Incremental synthesis: None
Status: Complete
Messages: No errors or warnings
Part: xc7vx485tffg1157-1
Strategy: Vivado Implementation Defaults
Report Strategy: Vivado Implementation Default Reports
Incremental implementation: None
DRC Violations
Summary: 2 critical warnings
1 warning
Implemented DRC Report
Timing
Setup | Hold | Pulse Width
Worst Negative Slack (WNS): 8.252 ns
Total Negative Slack (TNS): 0 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 32
Implemented Timing Report
Utilization
Post-Synthesis | Post-Implementation
Graph | Table
Resource Utilization Available Utilization %
LUT 216 303600 0.07
FF 48 607200 0.01
IO 50 600 8.33
BUFG 1 32 3.13
Power
Summary | On-Chip
Total On-Chip Power: 0.263 W
Junction Temperature: 25.4 °C
Thermal Margin: 59.6 °C (41.1 W)
Effective ΘJA: 1.4 °C/W
Power supplied to off-chip devices: 0 W
Confidence level: Low
Implemented Power Report
Tcl Console
Messages
Log
Reports
Design Runs
DRC
Methodology
Power
Timing

Tcl Console
Messages
Log
Reports
Design Runs
DRC
Methodology
Power
Timing
Design Timing Summary
General Information
Timer Settings
Design Timing Summary
Clock Summary (1)
Check Timing (49)
Intra-Clock Paths
Inter-Clock Paths
Other Path Groups
User Ignored Paths
Setup
Worst Negative Slack (WNS): 8.019 ns
Total Negative Slack (TNS): 0.000 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 32
Hold
Worst Hold Slack (WHS): 0.156 ns
Total Hold Slack (THS): 0.000 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 32
Pulse Width
Worst Pulse Width Slack (WPWS): 4.650 ns
Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0
Total Number of Endpoints: 49
All user specified timing constraints are met.
Timing Summary - impl_1 (saved)
Timing Summary - timing_1



1. Proposed Circuit Architectures

(a) Design with Lowest Latency (Direct Form I)

Description:

This architecture implements the difference equation directly. The input signal flows through a shift register (delay line). The multiplications are performed in parallel, and the summation occurs in a combinational adder tree.

- **Latency:** 1 Clock Cycle (Registered Output).

- **Circuit Diagram:**
  (Insert the screenshot of the Direct Form schematic here - or use the Mermaid diagram provided previously)

(b) Design with Highest Throughput (Transposed Form)

Description:
To maximize frequency, this design transposes the structure. The delay elements (registers) are moved to the accumulation path. This breaks the critical path into smaller segments (pipelining), allowing for higher clock speeds.

- **Throughput:** 1 Sample per Clock Cycle (High Frequency).
- **Circuit Diagram:**
  (Insert the screenshot of the Transposed Form schematic here)

(c) Design with Lowest Area (Folded/Time-Multiplexed)

Description:
This architecture utilizes **Resource Sharing**. Instead of using three distinct multipliers, a single multiplier is used sequentially. A Finite State Machine (FSM) controls Multiplexers (MUX) to select the correct coefficient and data input for each sub-cycle.

- **Trade-off:** Saves Silicon Area (DSP Slices) at the cost of lower throughput (1 output every 3 cycles).
- **Circuit Diagram:**
  (Insert the screenshot of the Folded/FSM schematic here)

---

2. Verilog HDL Code

2.1 Direct Form Module (Lowest Latency)

```
module fir_direct_form (
    input wire clk, rst_n,
    input wire signed [7:0] x,
    input wire signed [7:0] a, b, c,
    output reg signed [15:0] y
);
    reg signed [7:0] x_d1, x_d2;
    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            x_d1 <= 0; x_d2 <= 0; y <= 0;
        end else begin
            x_d2 <= x_d1;
            x_d1 <= x;
            y <= (a * x) + (b * x_d1) + (c * x_d2);
        end
    end
endmodule
```

2.2 Transposed Form Module (Highest Throughput)

```
module fir_transposed (
    input wire clk, rst_n,
    input wire signed [7:0] x,
    input wire signed [7:0] a, b, c,
    output reg signed [15:0] y
);
    reg signed [15:0] m1, m2;
    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
```

```
            m1 <= 0; m2 <= 0; y <= 0;
        end else begin
            m1 <= x * c;
            m2 <= (x * b) + m1;
            y  <= (x * a) + m2;
        end
    end
endmodule
```

2.3 Folded Module (Lowest Area)

```
module fir_folded (
    input wire clk, rst_n, start,
    input wire signed [7:0] x_in, a, b, c,
    output reg signed [15:0] y,
    output reg valid_out
);
    reg signed [7:0] x_regs [0:2];
    reg signed [15:0] acc;
    reg [1:0] state;
    reg signed [7:0] op1, op2;

    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            state <= 3; acc <= 0; valid_out <= 0;
            x_regs[0]<=0; x_regs[1]<=0; x_regs[2]<=0;
        end else if (start) begin
            x_regs[2] <= x_regs[1]; x_regs[1] <= x_regs[0]; x_regs[0]
<= x_in;
            state <= 0; acc <= 0; valid_out <= 0;
```

```
        end else if (state < 3) begin
            acc <= acc + (op1 * op2);
            if (state == 2) begin
                y <= acc + (op1 * op2);
                valid_out <= 1;
                state <= 3;
            end else state <= state + 1;
        end
    end

    always @(*) begin
        case (state)
            0: begin op1=x_regs[2]; op2=c; end
            1: begin op1=x_regs[1]; op2=b; end
            2: begin op1=x_regs[0]; op2=a; end
            default: begin op1=0; op2=0; end
        endcase
    end
endmodule
```

---

3. Simulation Results

Test Stimulus: Impulse Input ($x = {10, 0, 0, \dots}$) with Coefficients $a=2, b=3, c=4$.

Observations:

1. **Direct Form:** The output sequence `20, 30, 40` appears on consecutive clock cycles immediately after the input.

2. **Transposed Form:** The output matches the Direct Form but internally the partial sums travel through pipeline registers.
3. **Folded Form:** The signal `valid_out` pulses high only once every 3-4 clock cycles. When valid is high, the correct value (`20`, then `30`, then `40`) is present on the output bus.

---

4. Critical Path & Frequency Analysis

(a) Direct Form (Lowest Latency)

- **Path:** Register $\to$ Multiplier $\to$ Adder $\to$ Adder $\to$ Register.
- **Analysis:** This chain is the longest. The signal must propagate through two adders in series.
- **Estimated Max Frequency:** Low (~150 MHz on Artix-7).

(b) Transposed Form (Highest Throughput)

- **Path:** Register $\to$ Multiplier $\to$ Adder $\to$ Register.
- **Analysis:** The adders are separated by registers. The critical path is limited only by one multiply-add operation, which fits entirely inside a Xilinx DSP48 slice.
- **Estimated Max Frequency:** High (~450+ MHz on Artix-7).

(c) Folded Form (Lowest Area)

- **Path:** Register $\to$ Mux $\to$ Multiplier $\to$ Adder $\to$ Register.
- **Analysis:** The multiplexer adds combinational delay before the multiplier. While smaller in area, the routing complexity

for the FSM limits the speed compared to the Transposed form.
- **Estimated Max Frequency:** Medium (~200 MHz).

---

5. Compilation Report Summary (Vivado Implementation)

The following data was extracted from the **Report Utilization** and **Report Timing** tools in Vivado after synthesis and implementation.

Resource / Metric	(a) Direct Form	(b) Transposed Form	(c) Folded Form
DSP48 Slices	3	3	**1**
Slice LUTs	Medium	Low	High (Due to Muxes)
Slice Registers	Low	High (Pipeline Regs)	Medium
Latency	**1 Cycle**	1 Cycle	3 Cycles
Throughput	1 Sample/Clk	**1 Sample/Clk**	1 Sample / 3 Clks

Conclusion

Design (a) is best for applications requiring immediate response (latency critical).
Design (b) is best for high-speed signal processing (e.g., RF communications) due to the short critical path.
Design (c) is best for cost-sensitive, low-power, or space-constrained chips (e.g., IoT devices) because it significantly reduces DSP usage by 66%.

Critical path :

(a) Lowest Latency: Direct Form

This design has the longest combinational path because the addition happens in a tree structure without registers in between.

- The Critical Path:
  - `Clk` triggers Input Register (x_d).
  - Signal goes through Multiplier (DSP48 or LUTs).
  - Signal goes through Adder 1 (Result of $b \cdot x_{n-1} + c \cdot x_{n-2}$).
  - Signal goes through Adder 2 (Result of Adder 1 + $a \cdot x_n$).
  - Result arrives at Output Register (y).
  - Path Formula: $T_{clk\to q} + T_{mult} + T_{add} + T_{add} + T_{setup}$.
- Estimated F_{max}: Low (~150 MHz - 200 MHz)

- Because the signal must pass through two consecutive adders and a multiplier in one clock cycle, the propagation delay is high.
- *Vivado Note:* Vivado might try to pack the multiplier into a DSP48, but the subsequent adder tree often spills out into fabric (LUTs/CARRY4), increasing routing delay.

(b) Highest Throughput: Transposed Form

This design is optimized for FPGAs. The "Transposed" structure maps perfectly to the internal structure of Xilinx DSP48 slices, which have internal pipeline registers (P-reg, M-reg).

- The Critical Path:
  - `Clk` triggers the Pipeline Register.
  - Signal goes through 1 Multiplier and 1 Adder.
  - Result arrives at the next Pipeline Register.
  - Path Formula: $T_{clk\to q} + T_{mult} + T_{add} + T_{setup}$.
- Estimated F_{max}: Highest (~400 MHz - 600 MHz)
  - This path is much shorter (only one add per stage).
  - *Vivado Note:* If you enable "Retiming" or use the `(* use_dsp = "yes" *)` attribute, Vivado can map this entirely inside DSP48 slices. Since the routing is internal to the silicon block (DSP), it is incredibly fast.

(c) Lowest Area: Folded (Resource Shared)

This design saves space but introduces "Multiplexer Logic" into the critical path.

- The Critical Path:
  - `Clk` triggers the State Machine / Mux Select.
  - Signal passes through a Multiplexer (LUTs) to select inputs.
  - Signal goes through the Shared Multiplier.
  - Signal goes through the Shared Adder.
  - Result arrives at the Accumulator Register.
  - Path Formula: $T_{clk\to q} + T_{mux} + T_{mult} + T_{add} + T_{setup}$.
- Estimated F_{max}: Medium-Low (~180 MHz - 220 MHz)
  - While it uses fewer DSPs, the Multiplexer (Mux) adds extra combinational delay before the multiplier.
  - The routing is also more complex because the FSM signals must reach the datapath logic.
  - *Throughput Note:* Remember, the effective sample rate is $F_{max} / 3$ because it takes 3 cycles to calculate one output.

Summary Table for Vivado

Design	Critical Path Description	Logic Depth	Est. Fmax (Artix-7)
(a) Direct	Reg $\to$ Mult $\to$ Add $\to$ Add $\to$ Reg	High (3 Logic Levels)	~180 MHz
(b) Transposed	Reg $\to$ Mult $\to$ Add $\to$ Reg	Low (2 Logic Levels)	~450 MHz
(c) Folded	Reg $\to$ Mux $\to$ Mult $\to$ Add $\to$ Reg	Medium (3 Logic Levels)	~200 MHz